

1. Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

Step 1

- Network = **192.168.10.0/24**
- Need = **4 Subnets**

Borrow **2 bits**

$$2^2 = 4 \text{ Subnets}$$

Step 2

New Subnet Mask

$$/26 = 255.255.255.192$$

Subnets

Subnet	Network Address	Broadcast Address	Host Range
Subnet 1	192.168.10.0/26	192.168.10.63	192.168.10.1 – 192.168.10.62
Subnet 2	192.168.10.64/26	192.168.10.127	192.168.10.65 – 192.168.10.126
Subnet 3	192.168.10.128/26	192.168.10.191	192.168.10.129 – 192.168.10.190
Subnet 4	192.168.10.192/26	192.168.10.255	192.168.10.193 – 192.168.10.254

2. For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

Hint: Make sure not to use the network or broadcast address for any device.

Using Subnet 1

- Network Address = **192.168.10.0**
- Broadcast Address = **192.168.10.63**

Device	IP Address
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile	192.168.10.5

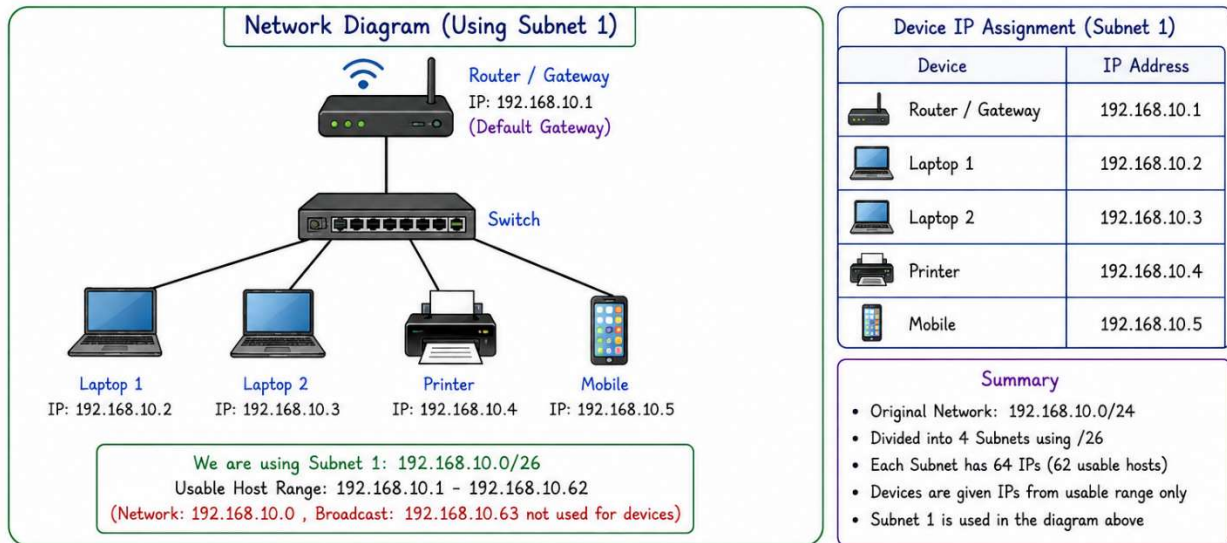
Subnet Mask: 255.255.255.192

Default Gateway: 192.168.10.1

3. Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.

Subnet Plan for 192.168.10.0/24 (Divided into 4 Subnets)

Subnet 1 - 192.168.10.0/26	Subnet 2 - 192.168.10.64/26	Subnet 3 - 192.168.10.128/26	Subnet 4 - 192.168.10.192/26
Network Address : 192.168.10.0	Network Address : 192.168.10.64	Network Address : 192.168.10.128	Network Address : 192.168.10.192
Broadcast Address : 192.168.10.63	Broadcast Address : 192.168.10.127	Broadcast Address : 192.168.10.191	Broadcast Address : 192.168.10.255
Usable Host Range: 192.168.10.1 - 192.168.10.62	Usable Host Range: 192.168.10.65 - 192.168.10.126	Usable Host Range: 192.168.10.129 - 192.168.10.190	Usable Host Range: 192.168.10.193 - 192.168.10.254



4.Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Step 1

We have the network:

192.168.10.0/24

We need **4 subnets**.

Step 2

Calculate subnet bits.

$$2^2 = 4$$

So, borrow **2 bits**.

Step 3

New subnet mask becomes:

$$/24 + 2 = /26$$

Subnet Mask = 255.255.255.192

Step 4

Find the number of hosts.

$$32 - 26 = 6 \text{ Host Bits}$$

$$2^6 = 64 \text{ IP Addresses}$$

$$64 - 2 = 62 \text{ Usable Hosts}$$

So each subnet has:

- **64 Total IP Addresses**
 - **62 Usable Host Addresses**
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Step 5

Choose the first subnet:

Network Address = 192.168.10.0

Broadcast Address = 192.168.10.63

Host Range = 192.168.10.1 – 192.168.10.62

Step 6

Assign IP addresses to devices.

Router : 192.168.10.1

Laptop 1 : 192.168.10.2

Laptop 2 : 192.168.10.3

Printer : 192.168.10.4

Mobile : 192.168.10.5

Final Summary

- Original Network: **192.168.10.0/24**
- New Subnet Mask: **255.255.255.192 (/26)**
- Total Subnets: **4**
- Usable Hosts per Subnet: **62**